

Vehicle Accident Alert System with Obstacle Detection

Project Overview

The **Vehicle Accident Alert System with Obstacle Detection** is an embedded safety solution designed to automatically detect vehicle accidents and instantly alert emergency contacts with the vehicle's real-time location.

The system also includes **IR-based obstacle detection** to provide short-range alerts and enhance driver safety.

This project integrates an **accelerometer, GPS module, GSM module, and IR sensor** using an Arduino-compatible microcontroller to create a reliable and low-cost accident monitoring system.

Objective

- Detect sudden vehicle impacts using an accelerometer
 - Automatically send **SMS alerts with GPS location**
 - Initiate an **emergency phone call**
 - Provide **IR-based obstacle alerts** during normal driving
 - Reduce emergency response time after road accidents
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System Architecture

The system continuously monitors acceleration values. When a sudden abnormal change is detected:

1. Accident is confirmed
2. GPS location is fetched
3. GSM module sends SMS & makes a call
4. Buzzer alerts locally

Obstacle detection works independently when no accident is detected.



Hardware Components

Component	Description
Arduino (UNO/Nano)	Main controller
ADXL345 Accelerometer	Impact detection
SIM800L GSM Module	SMS & calling
NEO-6M GPS Module	Location tracking
IR Obstacle Sensor	Short-range obstacle detection
Buzzer	Audio alert
Push Button	Manual system reset
Power Supply	5V regulated



Pin Configuration

GSM (SIM800L)

- RX → D3
- TX → D2

GPS (NEO-6M)

- RX → D9
- TX → D10

Sensors & Outputs

- IR Sensor → D6
 - Buzzer → D12
 - Reset Button → D11
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Software & Libraries Used

- Arduino IDE
 - TinyGPS++
 - Adafruit ADXL345 Library
 - Adafruit Unified Sensor Library
 - SoftwareSerial
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Working Principle

1 Impact Detection

- ADXL345 measures acceleration along X, Y, Z axes
- Sudden acceleration change magnitude is calculated
- If magnitude exceeds the defined sensitivity threshold → **Accident detected**

2 GPS Location Fetching

- GPS module attempts to lock satellite signal
- Latitude & longitude are extracted
- Google Maps link is generated

3 Emergency Alert System

- GSM module:
 - Makes a call to emergency contact
 - Sends SMS with live location link

4 IR Obstacle Detection

- IR sensor detects nearby obstacles

- Buzzer provides warning beeps
- Disabled automatically during accident handling

Manual Reset

- Push button resets alert state
 - Prevents false emergency alerts
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Alert Message Format

Accident Alert! Location: <http://maps.google.com/maps?q=loc> ;

Safety Features

- Automatic accident detection
 - Emergency call + SMS redundancy
 - Local buzzer alert
 - Manual reset option
 - Obstacle warning system
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Advantages

- Fully automatic emergency response
 - Low cost and compact design
 - Real-time GPS tracking
 - Modular and expandable
 - Suitable for two-wheelers and cars
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Limitations

- GPS may fail indoors or in tunnels

- GSM network dependency
 - Requires proper power regulation for SIM800L
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Future Enhancements

- IoT cloud integration
 - Mobile app for live tracking
 - Camera-based accident verification
 - Heartbeat/biometric monitoring
 - Speed-based accident severity analysis
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Applications

- Personal vehicle safety
 - Fleet management systems
 - Emergency response automation
 - Smart transportation systems
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Conclusion

This project demonstrates a practical implementation of an intelligent vehicle safety system using embedded electronics and communication modules. It effectively reduces emergency response time and enhances road safety using automation.